

The Promised Land?

Municipal Electoral Cycles and Land Consumption in Italy

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Motivation

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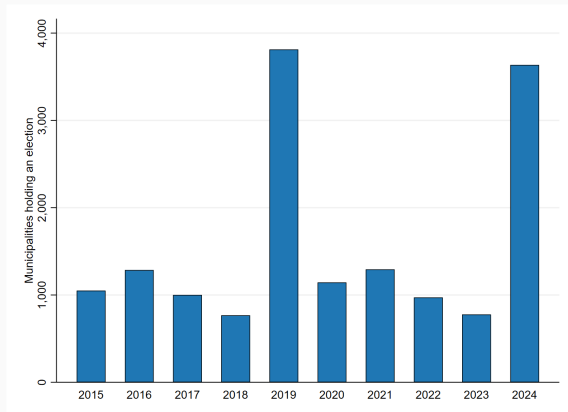
- Political Budget Cycle literature studies how politicians use fiscal or regulatory instruments to gain voters' preferences
- Italian evidence has focused mainly on fiscal instruments: capital expenditure, road investment (Repetto, 2018; Alesina et al., 2019; Ferraresi et al., 2019; Bracco et al., 2024; Giacobbe et al., 2024), but less on regulatory instruments
- Fiscal space has been shrinking: 2008 ICI abolition, and since 2016 a binding balanced-budget rule (*pareggio di bilancio*, L. 228/2015)
- Our hypothesis: mayors substitute toward regulatory instruments — what does granting a building permit cost the municipality? Nothing on the balance sheet, unlike spending or tax cuts

This Paper

- Italian mayors have direct control over land-use regulation — do they use it electorally, and if so, **before** the vote (signalling to persuade) or **after** (activating a mandate, repaying political debt)?
- We find a **post-election mandate cycle** in construction sites — and it reverses for lame-duck mayors, tying the mechanism to **future** re-election prospects, not the vote just held.

Setting & Data

Institutional Setting



Municipalities holding an election, by year (2015–2024)

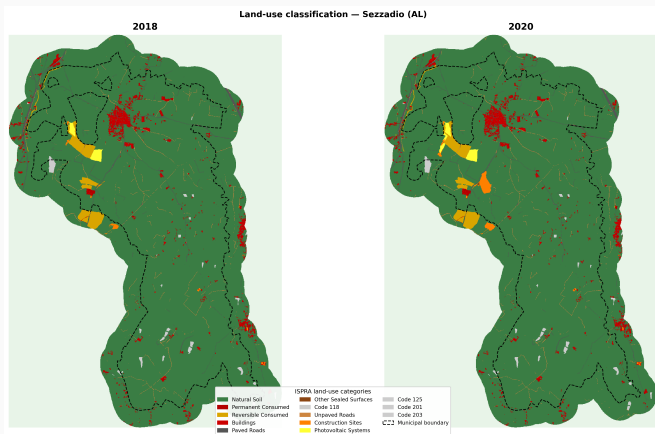
Elections data: Ministry of Interior, *Archivio Storico Elezioni Amministrative*.

- Mayors directly elected, 5-year term, **staggered** across municipalities
- Consecutive-term limits: 2 terms if pop. > 3,000; 3 terms if pop. ≤ 3,000
- Mayor controls both zoning (*Piano Regolatore*) and building permits (*permesso di costruire*)

Outcome Variable

- Ideally we would measure building permits directly, but these data are **not available at municipal level** in a systematic way.
- Instead we use the **area (m²) classified as a construction site (*cantieri*)** as the outcome variable from the ISPRA dataset about land consumption (*Carta Nazionale del Consumo di Suolo*): 10m satellite raster, annual, 2015–2024.
 - we use the most direct manifestation of activity: land is flagged as soon as it is cleared and fenced, **before any building is erected**
 - Buildings and roads register only at completion — a slow signal, and a natural placebo
 - A *cantiere* pixel originates from **non-consumed** land (agricultural/natural) — this measures genuine new land take (soil sealing), not reallocation of already-built surface

Measuring the Outcome



Sezzadio (AL), pop. 1,257 — 2019 election

2018 ($t-1$) vs. 2020 ($t+1$): new construction-site pixels (orange)

Empirical Strategy

The Estimation Puzzle

- To study dynamic effects, we use the **Event Study** design — the dynamic specification of Diff-in-Diff (**Miller, 2023**)
- Municipalities don't choose when to hold elections (**Coviello and Gagliarducci, 2017**), which rules out selection bias — but timing still matters: being in (or out of) an election year affects both treated and control units
- Two problems remain: **staggered timing** makes some comparisons forbidden (**Goodman-Bacon, 2021**), and elections **recur** every 5 years — a pulse, not an absorbing treatment — so **Callaway and Sant'Anna (2021)**; **Sun and Abraham (2021)** do not apply

Solution: Stacked Event Study (1/3)

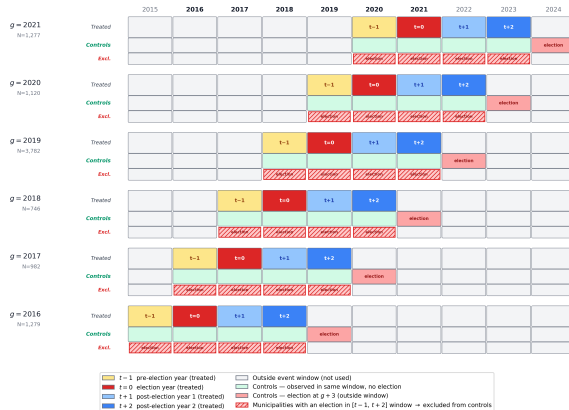
Stacked Event Study (Baker et al., 2022; Wing et al., 2024): a separate dynamic DiD “experiment” for each election cohort, each with its own clean control group, then pooled together

- Elections recur every 5 years — the same municipality can elect in more than one candidate cohort year (e.g., both $g = 2016$ and $g = 2021$)
- For each stack — a sub-sample defined by cohort year $g \in \{2016, \dots, 2021\}$: municipality i is treated ($D = 1$) if its own election year equals g ; municipality j is a control ($D = 0$) if its own election year equals $g + 3$ (we decided the window).
- Because elections recur, the same municipality can be treated in one stack and control (or treated again) in another
- Feasibility ($g - 1 \geq 2015$, $g + 3 \leq 2024$) $\Rightarrow$ only $g \in \{2016, \dots, 2021\}$ survives.

Solution: Stacked Event Study (2/3)

Stacked DiD Design — Cohort Structure and Clean Control Groups

Panel B — Event-window calendar: treated, eligible controls, and excluded municipalities



Solution: Stacked Event Study (3/3)

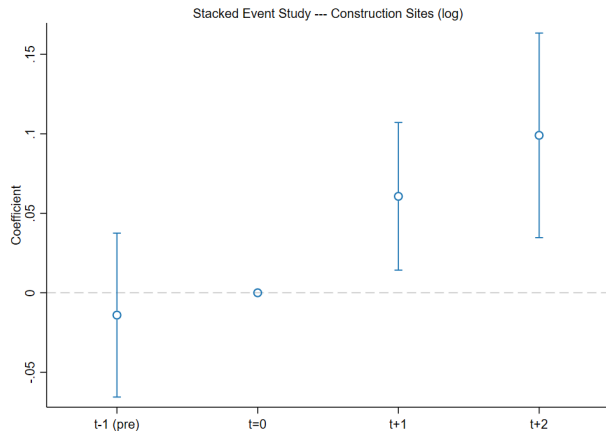
Practically, we create a dummy for each stack and estimate the following regression:

$$Y_{it}^s = \sum_{\tau \in \{-1, +1, +2\}} \delta_{\tau} D_i^s \cdot \mathbf{1}[t = \tau] + \mathbf{x}_{it}'\gamma + \mu_i^s + \nu_t^s + \varepsilon_{it}^s$$

- s : stack index — the same cohort g as before, now written s $t = y - g$: event time baseline $t = 0$
- $D_i^s = 1$ if municipality i is treated in stack s , 0 if control
- $\mathbf{x}_{it}$: controls — log population, gender, age at election, education, electoral system, party bloc (left, right)
- stack FE: stack $\times$ municipality, stack $\times$ year
- SE clustered at municipality level; $Y = \ln(\text{construction-site area, m}^2 + 1)$
- Estimated by [WLS](#), not OLS ([Wing et al., 2024](#)): $\omega_i^s = 1$ if treated, $\omega_i^s = N_D^s / N_C^s$ if control — corrects for unbalanced stack sizes

Results

Main Result



Flat before the election, surging after

$$D_{-1} = -0.014 \text{ (n.s.)}$$

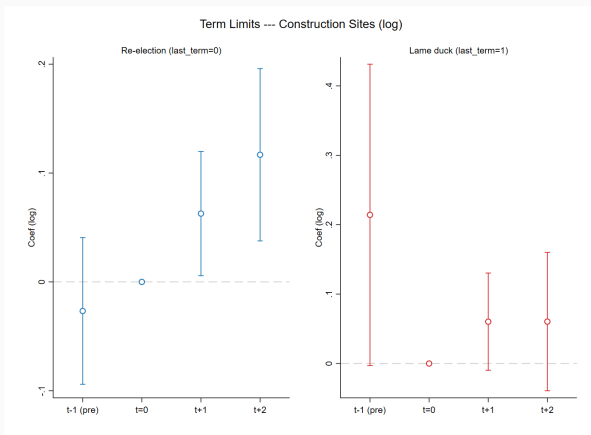
$$D_{+1} = 0.061^{**}$$

$$D_{+2} = 0.099^{***}$$

[full table →](#)

[TWFE benchmark →](#)

Mechanism: The Re-election Incentive



Left: re-election eligible. Right: term-limited (lame duck).

The surge is about re-election, not a fixed administrative lag

- Eligible mayors: flat pre-election, surge after
- Lame ducks: $D_{-1} = +0.214^*$ — reverses

Testing the Mechanism: permit Timing (1/2)

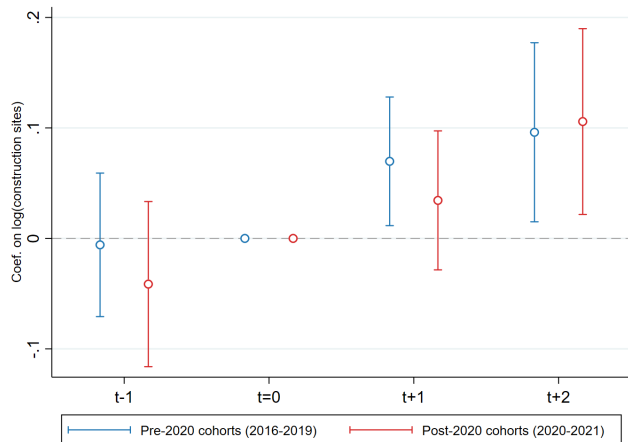
Alternative story: the permit was granted *before* the election, and we're just picking up a delayed construction start

- **The rule** (Art. 15 DPR 380/2001): works must start within 1 year of permit issuance; lapse is automatic, and municipal rules cannot override this national term (Corte Cost. 245/2021)
- **Bound hypothesis:** a permit granted any time up to the election can legally be deferred to *at most* $t+1 \Rightarrow$ a mechanical delay cannot produce a significant effect at $t+2$
- **Shape hypothesis:** a stock of pre-election permits being mechanically flushed out should *decay* ($D_{+1} > D_{+2}$) — we instead see it *grow* ($D_{+2} > D_{+1}$)
- **Complication:** post-2020 extensions (DL 76/2020 on) loosen the 1-yr bound — tested directly $\rightarrow$ next slide

Testing the Mechanism: permit Timing (2/2)

Testing the “permit signed before, site opens late” alternative

Pre-2020 (N=50,940) vs. post-2020 (N=25,192) cohorts



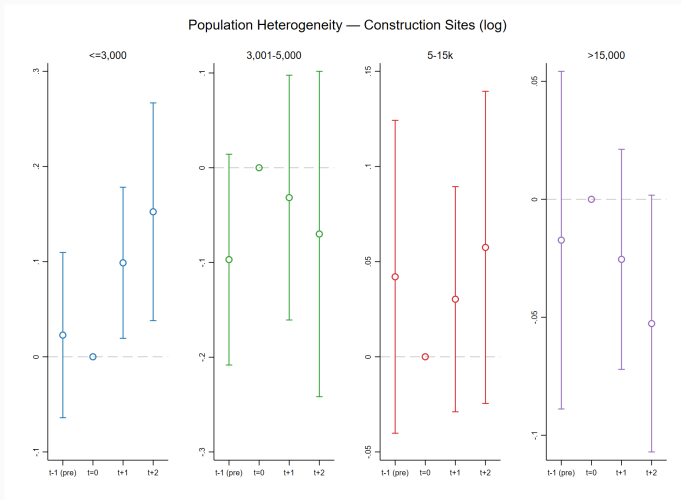
How Big Is the Effect?

Re-normalised to the pre-election baseline

- +8% at $t+1$, +12% at $t+2$ relative to $t-1$
- Median municipality: $\approx 950 \text{ m}^2$ of new construction-site area at $t+2$
⇒ footprint of a small residential plot

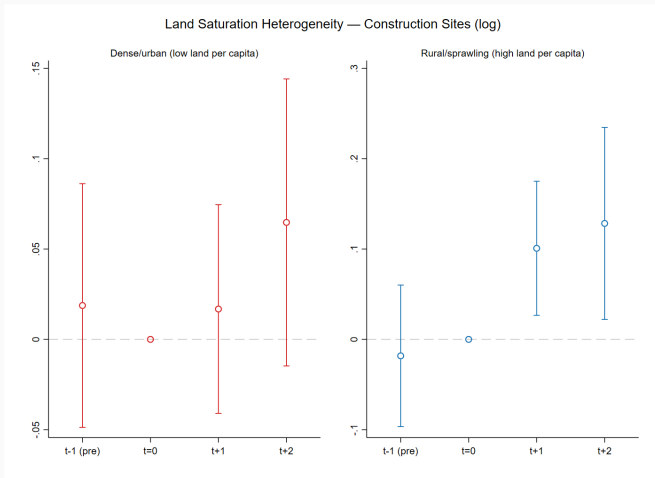
Heterogeneity & Robustness

Who Drives the Cycle? Size



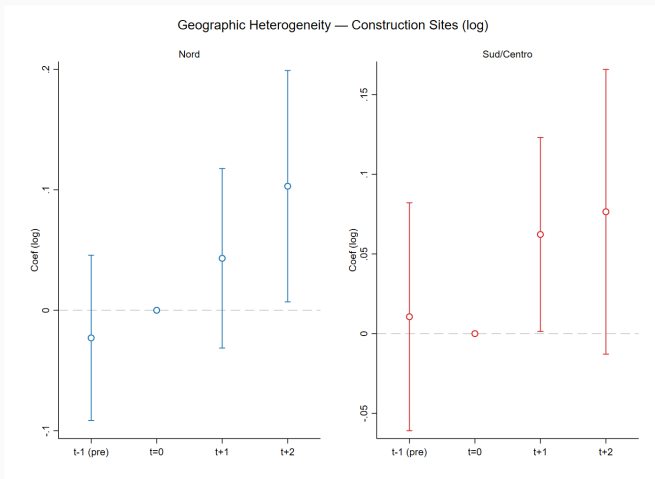
The Cycle lives in municipalities $\leq 3,000$ inhabitants; null above

Who Drives the Cycle? Rurality



Stronger in rural areas, where there is more land per-capita

Who Drives the Cycle? Geography



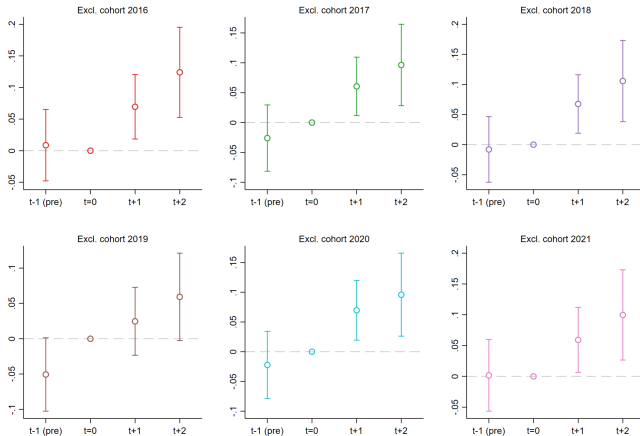
Stronger in the North

North: $D_{+2} = 0.103^{**}$.

Centre-South: $D_{+2} = 0.076^*$, less precise.

Robustness

Cohort-exclusion robustness --- Construction sites (log)



Not an artefact of any single election cohort

Each panel excludes one election cohort in turn

parallel-trends →

Conclusion

Conclusion

Post-election mandate cycle in construction sites in municipalities with $\leq 3,000$ inhabitants — not a mechanical lag — in the north and rural areas when mayors face a re-election incentive.

- Fiscal compression $\Rightarrow$ substitution toward regulatory channels
- This is **not a neutral timing shift**: *cantieri* originate from non-consumed land, so the cycle accelerates **soil sealing** — independent of demographic need, since population is held fixed
- **Policy**: soil-consumption caps (*consumo di suolo* legislation, EU no-net-land-take 2030) should account for this electoral-timing channel — targeting *when* permits are granted (disclosure near elections) preserves local accountability better than centralizing permit authority

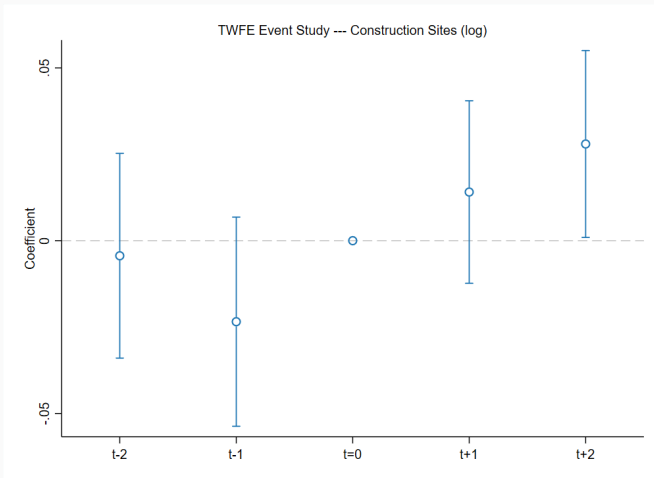
Thank you

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Backup: TWFE Benchmark



Same sign/significance as Stacked ES, smaller magnitude

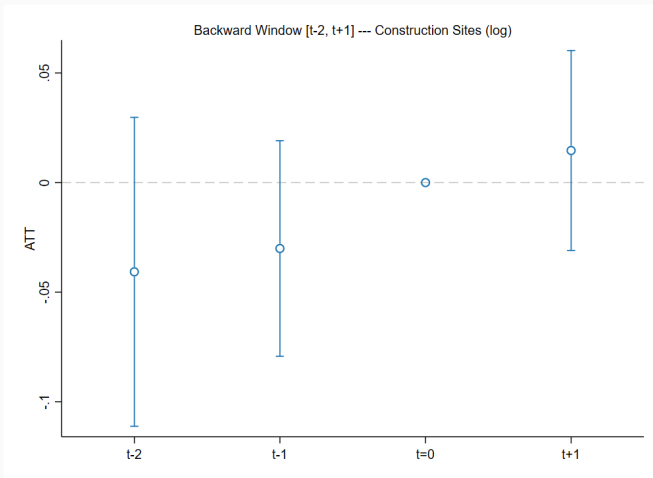
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Backup: Full Regression Table

	Construction sites (log)	
D_{-1}	-0.0140	(0.0263)
D_{+1}	0.0607**	(0.0237)
D_{+2}	0.0990***	(0.0328)
In Population	-1.6234***	
Female mayor	-0.0527	
Mayor age	-0.0023	
University degree	-0.0630*	
N	75,908	

[← back](#)

Backup: Parallel Trends (Backward Window)



$t - 2$ coefficient: -0.041 (s.e. = 0.036), statistically null

← back

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